

Project Initialization and Planning Phase

Date	04 July 2026
Team ID	SWTID-2026-7391
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	To develop an AI and Machine Learning-based Credit Card Approval Prediction System that analyzes customer information such as income, employment status, credit score, age, and financial history to predict whether a credit card application should be approved or rejected.
Scope	The project focuses on building an intelligent prediction system for credit card approval. It includes customer data collection, data preprocessing, machine learning model training, prediction, and result visualization.
Problem Statement	
Description	Traditional credit card approval methods rely on manual verification, which is time-consuming, error-prone, and inconsistent. Banks often need to analyze multiple customer attributes such as income, employment, credit history, and existing liabilities before making a decision. This process delays approvals and affects customer experience.
Impact	An automated AI-based prediction system improves the speed and accuracy of credit card approval decisions. It reduces manual effort, minimizes human errors, enhances customer satisfaction, and helps financial institutions make reliable and data-driven decisions.
Proposed Solution	
Approach	Develop a Machine Learning model that analyzes customer details, financial information, employment status, and credit history to predict credit card approval. The system processes the input data, validates it, performs prediction using trained ML algorithms, and displays an instant approval or rejection result.
Key Features	AI & Machine Learning-based prediction Fast and accurate approval decisions Secure customer data processing

	User-friendly web interface Real-time prediction results Reduced manual verification Reduced manual verification
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processor required to train and run the ML model.	Intel i5/Ryzen 5 Processor
Memory	RAM required for data processing and model execution.	8 GB RAM
Storage	Disk space used to store datasets, models, and project files.	256 GB SSD or above
Software		
Frameworks	Backend framework used to build the web application	Flask
Libraries	Libraries used for data processing, model training, and prediction	scikit-learn, pandas, numpy, matplotlib, Joblib
Development Environment	Tools used to write, test, and run the project.	Jupyter Notebook, VS code
Data		
Data	Source of customer application and credit-related data used for training and testing the ML mode	Credit Card Approval Dataset (CSV format)